

PENNSYLVANIA MEASLES OUTBREAK ANALYSIS

Vaccination Rate and Population Density as Predictors of the 2026 Pennsylvania Measles Outbreak: A County-Level Analysis

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Abstract

Pennsylvania is experiencing its largest measles outbreak in decades, raising the question of whether vaccination coverage, population density, or both are the primary structural predictors of county-level transmission. County-level case counts (August 17 to September 16, 2026) were analyzed against kindergarten MMR vaccination rates and population density using Pearson, Spearman, and Kendall correlations; negative binomial and zero-inflated negative binomial regression; and global and local Moran's I spatial autocorrelation. Vaccination rate was a significant, stable negative predictor of case counts across the study period ($\beta = -23.88$, $p = 0.003$), corresponding to a 21.2% reduction in case count per percentage-point increase in coverage. Density showed no meaningful raw correlation with cases but became significantly negative once vaccination rate was held constant ($\beta = -0.002$, $p = 0.002$), a suppression effect reflecting that denser counties tend to have higher vaccination coverage. Global Moran's I confirmed significant spatial clustering ($I = 0.156$, $p < 0.0001$) concentrated in Lancaster, Chester, Lebanon, and York Counties, with Berks and Dauphin as significant negative outliers despite direct geographic adjacency to the cluster, a pattern inconsistent with simple geographic contagion. A Wilcoxon rank-sum test confirmed that Lancaster's under-vaccination is concentrated within a socially bounded subgroup: plain-community schools (median MMR = 57.1%) had significantly lower coverage than all other schools in the county (median = 91.4%, $W = 472$, $p < 0.001$). Comparison of Lancaster, Allegheny, and Mifflin Counties revealed two distinct structural pathways capable of sustaining a large outbreak once seeded: a concentrated, socially bounded under-vaccinated subgroup, or a diffuse, county-wide vaccination gap. A comparable under-vaccinated pocket in Allegheny County has produced zero cases absent an introduction event. These findings indicate that vaccination structure, not density or raw geographic proximity, is the primary driver of outbreak spread, and that surveillance should not rely on a single visible marker of risk.

Keywords: measles, vaccination rate, population density, spatial autocorrelation, negative binomial regression, outbreak surveillance

Introduction

The United States is on track to lose its measles elimination status, one that has been held since 2000, with large outbreaks in several states becoming the norm over the last few years. Measles is the most contagious infectious disease, with an R_0 between 12 and 18, meaning that each infected individual is expected to infect 12 to 18 additional individuals (Guerra et al., 2017). Official data from the CDC shows that continuous domestic transmission has lasted well beyond the 12-month threshold required for disease elimination (CDC, 2026). Pennsylvania is currently seeing its largest measles outbreak in decades, with four deaths as of writing on September 17, 2026, the most in the US since this elimination status was reached. As of this date, there are over 700 cases in the state (Pennsylvania Department of Health, n.d. -a) (Figure 1).

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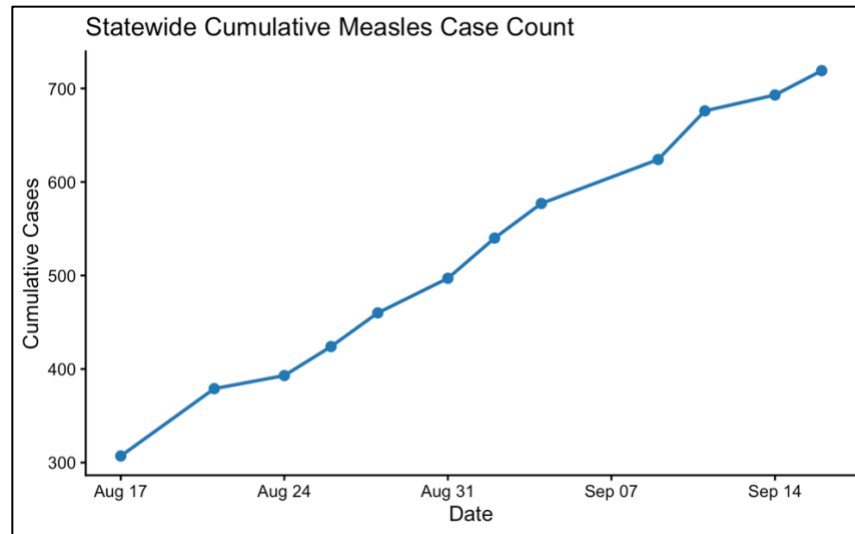


Figure 1

The mortality rate of this outbreak is approximately 0.5% (4/731) in this outbreak, while the true mortality rate of measles is roughly 0.1% in the developed world (Cairns et al., 2010). This misalignment suggests that the true case count in Pennsylvania is likely substantially greater.

Data and Methods

Data Sources

Data for this study were collected from multiple sources. Measles case counts were pulled with each update on Monday, Wednesday, and Friday from the Pennsylvania Department of Health Measles Dashboard (Pennsylvania Department of Health, n.d. -a). Pennsylvania school and county immunization rates were pulled from the Pennsylvania School Immunization Rates Interactive Data Tool (Pennsylvania Department of Health, n.d. -b). Pennsylvania population density was calculated by dividing population, pulled from the U.S. Census Bureau (U.S. Census Bureau, 2026), by area, pulled from the National Association of Counties (National Association of Counties, 2007).

Methods

Correlation was examined longitudinally between vaccination rate vs. cases and density vs. cases, and cross-sectionally between density vs. vaccination rate. Three different correlation measures were used to evaluate these trends: Pearson, Spearman, and Kendall. Pearson measures the strength and direction of the linear relationship, Spearman measures the strength and direction of the monotonic relationship based on differences in ranks, and Kendall measures the strength and direction of the monotonic relationship based on concordance of pairs.

Various models were analyzed to determine which model architecture was ideal for analyzing this relationship between density and vaccination rate. A Poisson model was analyzed and had statistically significant predictors but was ultimately not used due to its poor fit to

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overdispersion. A negative binomial model and zero-inflated negative binomial model was fit to handle this overdispersion and compared using Akaike Information Criterion (AIC). The zero-inflated model initially performed better due to only a small number of counties having cases, but as the outbreak expanded to more counties, the standard model became a consistent better fit (Figure 2).

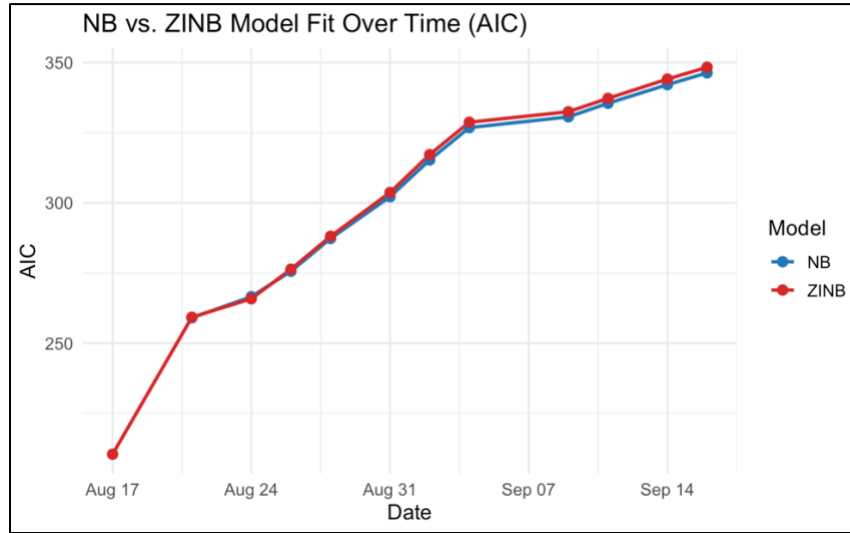


Figure 2

A negative binomial and zero-inflated negative binomial model were fit with an addition of mean religious exemption and the standard deviation of this count was also tested, but not used as all variables except for density were statistically insignificant.

Negative Binomial

$$\log(\mu_i) = \beta_0 + \beta_1 \cdot \text{Vaccination Rate}_i + \beta_2 \cdot \text{Density}_i + \log(\text{Population}_i)$$

$$Y_i \sim \text{NB}(\mu_i, \theta)$$

Zero-Inflated Negative Binomial

Count component:

$$\log(\mu_i) = \beta_0 + \beta_1 \cdot \text{Vaccination Rate}_i + \beta_2 \cdot \text{Density}_i + \log(\text{Population}_i)$$

Zero-inflation component:

$$\text{logit}(\pi_i) = \gamma_0$$

Combined likelihood:

$$\Pr(Y_i = 0) = \pi_i + (1 - \pi_i) \cdot f_{\text{NB}}(0 \mid \mu_i, \theta)$$

$$\Pr(Y_i = y_i) = (1 - \pi_i) \cdot f_{\text{NB}}(y_i \mid \mu_i, \theta), y_i > 0$$

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To test whether cases spread through ordinary geographic contagion, as opposed to spreading within a socially defined but geographically dispersed population, spatial autocorrelation was assessed using Moran's I. An adjacency matrix was built from Pennsylvania county boundaries, and Moran's I was calculated on case counts under this neighbor structure.

Results

Across all three correlation methods, density and vaccination rate is positively correlated across all three methods (Table 1).

Correlation: Density vs. Vaccination Rate	
Pearson	0.146
Spearman	0.226
Kendall	0.181

Table 1

Considering density vs. cases, the Pearson correlation stays near zero and consistent throughout, but Spearman and Kendall correlations are positive at the beginning of my study period and become negative as the outbreak progresses. This tells us that the sign-rank relationship is more sensitive to the outbreak expansion than linear methods. Considering vaccination rate vs. cases, the correlation was negative and relatively stable in sign and magnitude across all three methods (Figure 3).

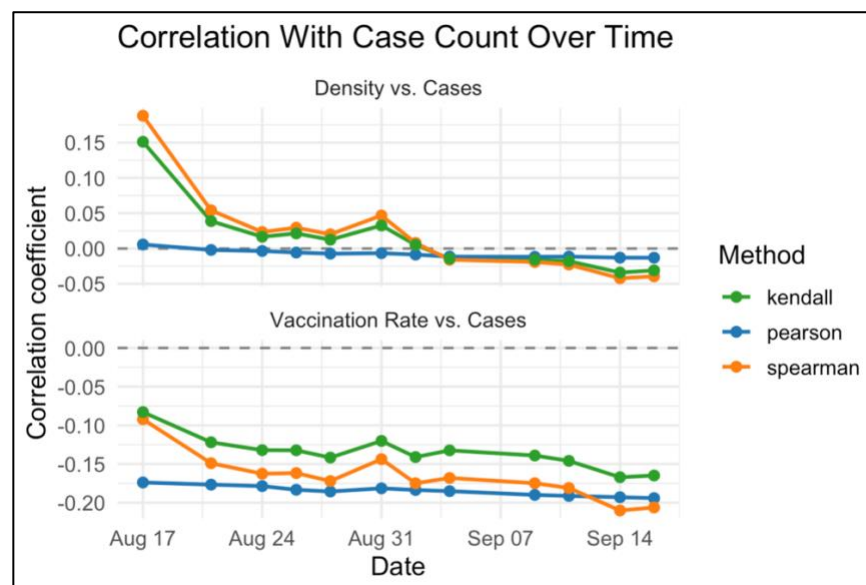


Figure 3

Figure 4 displays case count trajectories for the twelve highest-burden counties over the study period, showing Lancaster's sustained lead alongside Mifflin's and Chester's steeper, more recent emergent growth.

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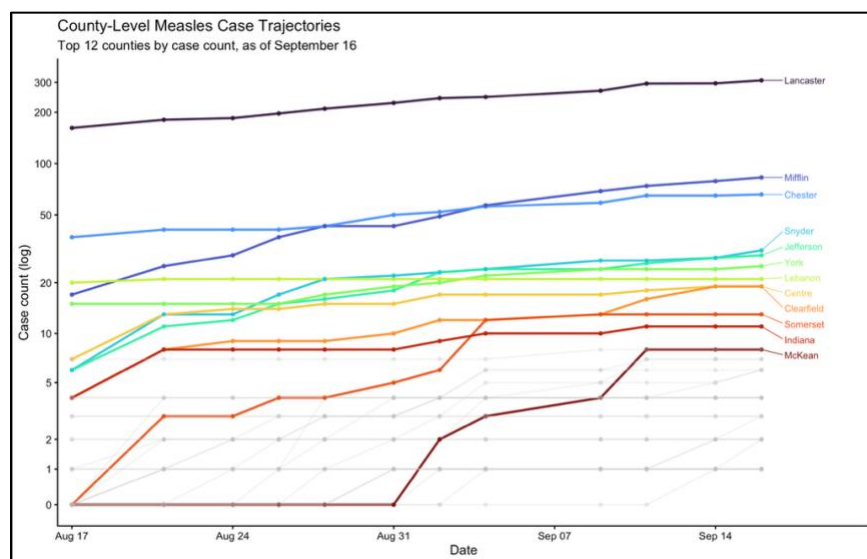


Figure 4

The estimated association between vaccination coverage and cumulative cases was directionally consistent across successive outbreak snapshots of the negative binomial model, while the density coefficient grew increasingly negative as the outbreak expanded to additional counties (Figure 5).

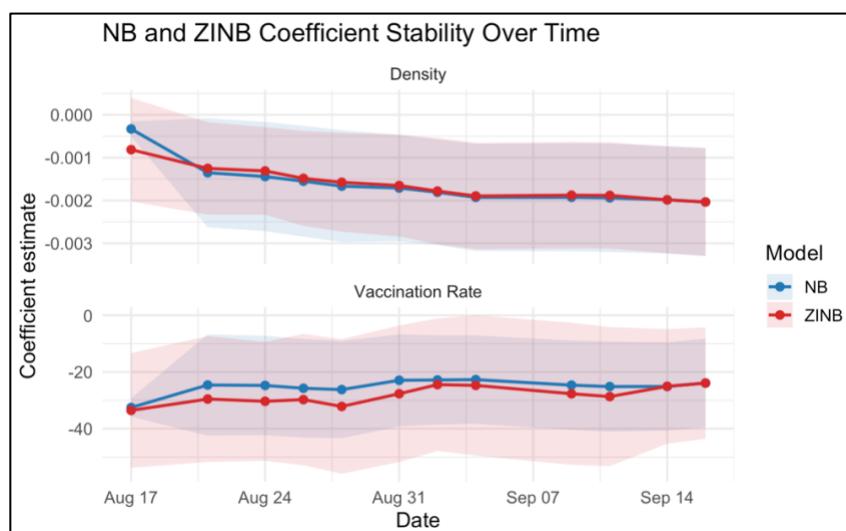


Figure 5

The final negative binomial model, fit using data through September 16, 2026, produced the coefficients shown in Table 2.

Term	Estimate	SE	z	p
Intercept	13.059	7.301	1.789	0.0737
Vaccination Rate	-23.878	7.964	-2.998	0.00272
Density	-0.00204	0.000646	-3.149	0.00164

Table 2

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These coefficients can be interpreted in the following manners:

- Each 1 percentage point increase in MMR vaccination rate is associated with a 21.2% decrease in case count, holding density constant.

$$e^{-23.878 \times 0.01} = 0.788$$

- Each 1 person per square mile increase in density is associated with a 0.2% decrease in case count, holding vaccination rate constant.

$$e^{-0.00204 \times 1} = 0.998$$

- Each 100 person per square mile increase in density is associated with a 18.5% decrease in case count, holding vaccination rate constant.

$$e^{-0.00204 \times 100} = 0.815$$

Global Moran's I was significantly positive ($I = 0.156$, $p < 0.0001$), indicating that cases are spatially clustered rather than randomly distributed across the state. Counties near other high-case counties tend to also have elevated case counts, more than chance alone would produce (Figure 6).

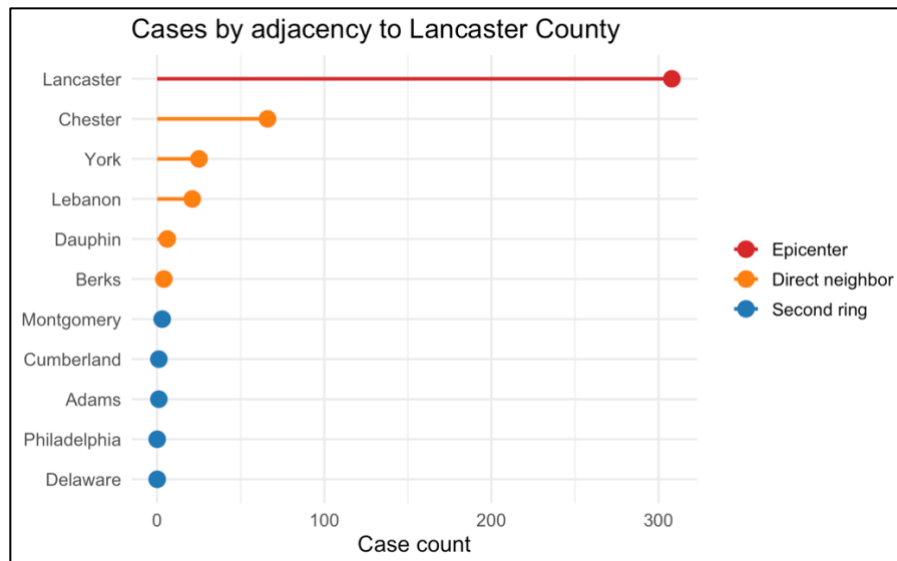


Figure 6

Local Moran's I was then used to identify which specific counties around Lancaster were driving this clustering (Table 3).

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County	Case Count	Local I	p
Lancaster	308	2.650	0.003
Chester	66	2.452	< 0.001
York	25	0.635	< 0.001
Lebanon	21	0.461	< 0.001
Dauphin	6	-0.108	0.008
Berks	4	-0.246	< 0.001

Table 3

Lancaster, Chester, Lebanon, and York form a significant high-high cluster; these counties are not just individually elevated, but are surrounded by other elevated counties, consistent with real spatial concentration around the epicenter. By contrast, Dauphin and Berks are significant negative outliers. Despite being geographically adjacent to Lancaster County, both counties recorded fewer cases than their neighbors would predict.

Upon examining the absolute size of severe pockets of under-vaccination (<70% coverage), it was revealed that Allegheny County, not Lancaster, had the largest number of children in severely under-vaccinated schools (Figure 7). Additionally, Allegheny (41.7%) and Lancaster (26.1%), as well as a handful of other counties, have extremely severe pockets of under-vaccination (Figure 8).

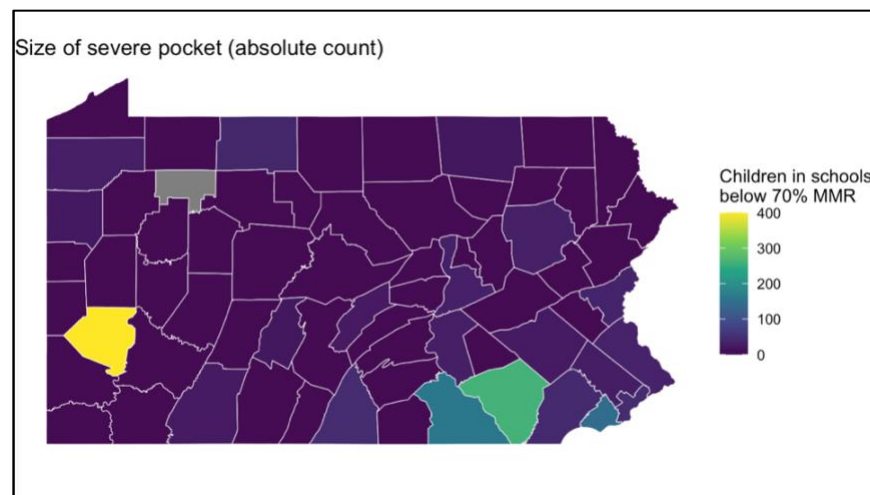


Figure 7

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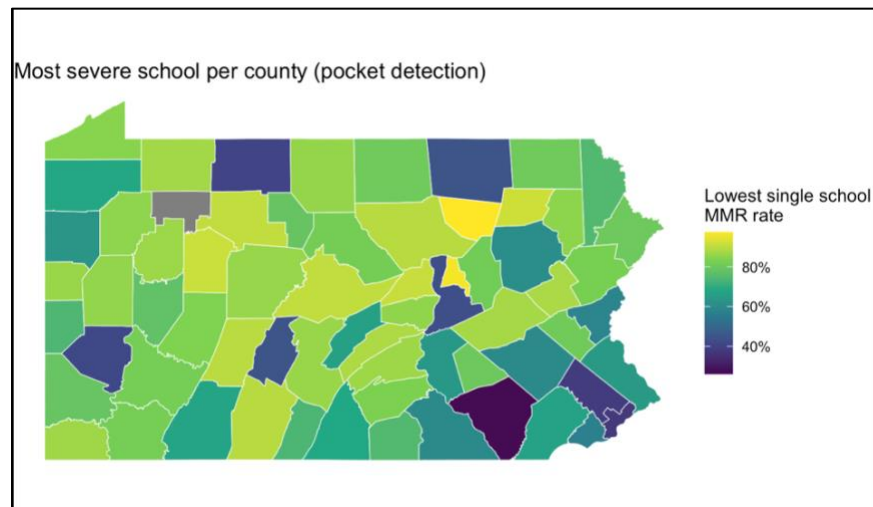


Figure 8

To determine whether Lancaster's under-vaccinated pocket truly reflects a distinct, socially bounded subgroup rather than generally depressed coverage across the county's schools, kindergarten MMR vaccination rates were compared between plain-community schools and all other schools in Lancaster County using a Wilcoxon rank-sum test. Plain-community schools ($n = 7$, median MMR rate = 57.1%) has significantly lower vaccination rates than all other schools in the county ($n = 76$, median MMR rate = 91.4%) $W = 472$, $p = 0.00076$ (Figure 9).

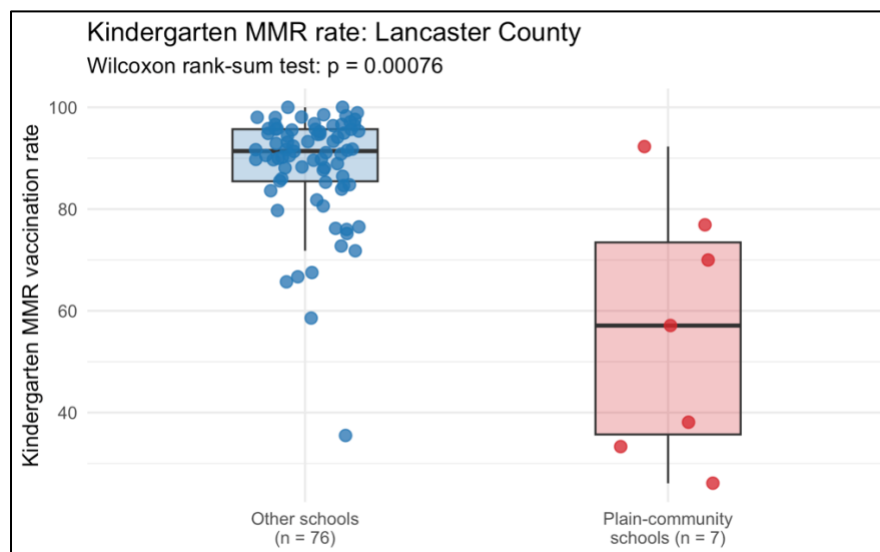


Figure 9

This confirms that Lancaster's severe under-vaccination pocket is concentrated within an identifiable community rather than being a broad trend across the entire county. This is a structural feature that distinguishes it from the diffuse, county-wide gap later observed in Mifflin County.

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At the time of writing, Allegheny County does not have any measles cases. Mifflin County now has the highest rate of cases in the state but lacks a severe under-vaccination pocket (Figure 10).

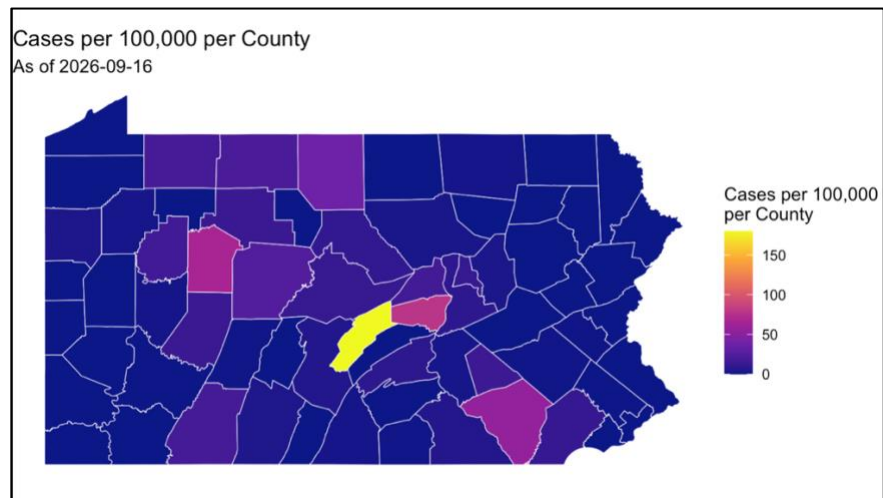


Figure 10

Mifflin County has 100% of kindergarteners in schools below the necessary herd immunity threshold (Figure 11).

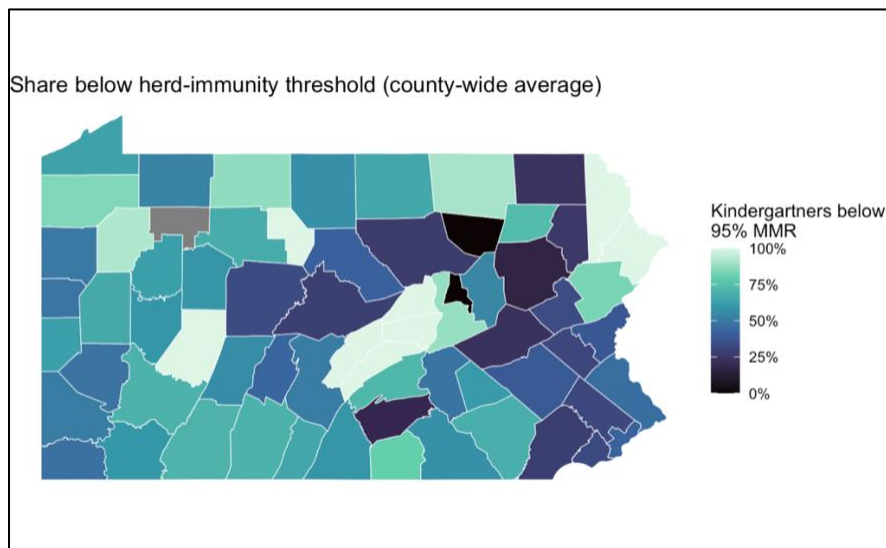


Figure 11

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Discussion

Vaccination rate proved to be a significant and consistent negative predictor of case count throughout the study period. This finding is intuitive: counties with higher vaccination coverage are less likely to sustain measles transmission. Density presented a more counterintuitive result. A naïve assumption might expect denser counties to experience more cases, given greater opportunity for person-to-person contact and transmission. Instead, the opposite relationship emerged once vaccination rate was held constant. This is a suppression effect. Since denser counties in this dataset tend to have higher vaccination coverage, the raw correlation between density and cases obscures a real negative relationship that only becomes visible in the multivariate model. This finding argues against density as an independent predictor of measles spread and instead points to vaccination rate as the primary structural factor underlying case transmission. The spatial clustering results reinforce this conclusion.

Figure 12 visualizes this relationship directly, as high-case counties cluster below the 95% herd-immunity threshold regardless of their position on the density axis, while several high-density counties above the threshold show minimal case counts.

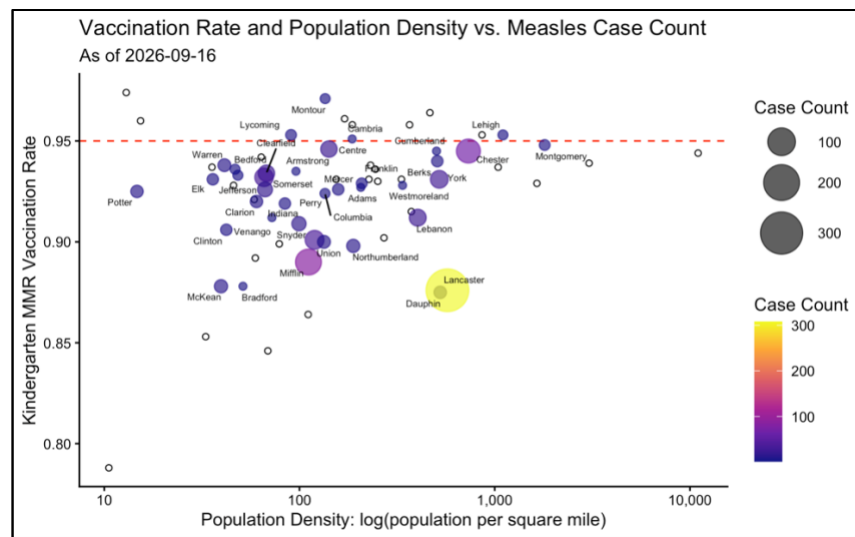


Figure 12

If the outbreak were spreading primarily through ordinary contagious contact across county lines, Berks County, directly adjacent to both Lancaster and Chester, would be expected to show elevated case counts of its own. Instead, Berks emerged as a significant negative outlier despite this adjacency. This pattern is inconsistent with simple geographic diffusion; it instead suggests that the outbreak tracks the boundaries of specific under-vaccinated communities rather than geographic proximity itself. Geography alone cannot explain why an adjacent county remained unaffected, but vaccination structure can.

The comparison between Lancaster, Allegheny, and Mifflin Counties refines this picture further. Allegheny County's severely under-vaccinated pocket is comparable in size to Lancaster's, yet

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Allegheny has recorded zero cases. This indicates that a large, concentrated under-vaccinated subgroup is necessary but not sufficient to sustain an outbreak; an introduction event is also required, which is unsurprising, since measles cannot manifest without a transmission event. Mifflin County's experience complicates the picture in the other direction. Rather than a concentrated pocket, Mifflin's entire kindergarten population sits below the herd immunity threshold, with no known internal subgroup contrast at all. Despite lacking Lancaster's characteristic concentrated-pocket structure, Mifflin now has the highest case rate in the state, confirming a second, independently sufficient pathway. Mifflin County indicates that a diffuse, county-wide vaccination gap in which the entire population of interest is susceptible, so that once a case is seeded, transmission can proceed readily.

Together, the Lancaster and Mifflin cases indicate that no single structural signature of under-vaccination is either necessary or sufficient on its own to produce a large outbreak. Lancaster illustrates a concentrated-pocket mechanism, in which a small, socially bounded subgroup sits far below herd immunity while the rest of the county remains close to it, and this subgroup alone is large enough to sustain transmission once introduced. Mifflin illustrates a diffuse mechanism, in which the entire county sits uniformly below threshold with no internal contrast. Both mechanisms reduce to the same underlying condition: a population of susceptible individuals large enough to sustain chains of transmission, combined with an introduction event. This reframes the original hypothesis. Rather than treating a large, visible under-vaccinated pocket as the single necessary marker of outbreak risk, the data support a typology of at least two distinct risk profiles, concentrated-minority gaps and diffuse county-wide gaps, that appear to carry comparable outbreak potential once seeded. From a surveillance standpoint, this argues against over-indexing on visible structural outliers as the primary risk signal, since counties with unremarkable, evenly distributed coverage below threshold may be an equally viable substrate for sustained transmission.

Taken together, these results answer the question posed at the outset of this analysis, that vaccination rate, not population density, is the primary structural predictor of Pennsylvania's ongoing measles outbreak, and the specific structure of under-vaccination, whether concentrated or diffuse, determines where and how readily that outbreak can take hold.

Limitations

A notable limitation of this analysis is that publicly available data do not report case locations at a resolution finer than the county level. This is a meaningful gap given the paper's central finding. The suppression effect and the Lancaster clustering result both point toward transmission occurring within a specific, geographically dispersed sub-population rather than uniformly across a county's general population. County-level case counts cannot distinguish between these two scenarios directly; a case attributed to Lancaster County is recorded identically whether it occurred within a plain-community enclave or in the downtown area, even though the underlying transmission dynamics are likely very different. This gap exists primarily

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because the Pennsylvania Department of Health, like most state health departments, applies small-cell privacy suppression standards to its public case data.

This creates a version of the modifiable areal unit problem (Openshaw, 1984). County-level density is a population-weighted average across the entire county, but if cases are concentrated in specific low-density rural pockets within an otherwise moderate-density county, the county-wide density figure does not reflect the density actually experienced by the affected population. The same issue applies to vaccination rate; the county average blends Plain-community schools with all other schools into a single number, which is precisely why the school-level analysis in the Introduction was necessary to surface a pattern that county-level regression alone could not detect. Absent case-level or sub-county geographic data, such as zip code or census tract, it is not possible to confirm directly that reported cases are disproportionately originating from these lower-vaccination sub-areas rather than being independently distributed across the county. Kindergarten MMR vaccine coverage was treated as a proxy for underlying population susceptibility rather than a contemporaneous measure of vaccination coverage among all age groups represented.

A second limitation concerns the temporal structure of the data. Each of the twelve snapshots analyzed represents cumulative case counts as of that date rather than independent incidence counts, and successive snapshots describe the same ongoing outbreak rather than separate, independent events. As a result, the consistency of the vaccination rate coefficient across snapshots should be interpreted as a robustness check on a single evolving outbreak, not as replication across independent samples.

Finally, as with the rest of this analysis, these are county- and school-level aggregate patterns describing population-level susceptibility structure, not individual-level risk, and should not be read as claims about any specific child's or family's vaccination status.

Conclusion

This analysis set out to answer a single question: is Pennsylvania's ongoing measles outbreak driven primarily by vaccination rate, population density, or both? The evidence assembled across each method points toward a consistent answer, with an important refinement uncovered along the way.

Vaccination rate emerged as the dominant and most stable predictor of case counts throughout the study period, remaining statistically significant across every snapshot of the negative binomial model. Population density, by contrast, showed no meaningful raw relationship with case counts, but revealed a suppression effect once vaccination rate was controlled for. Denser counties trend toward higher vaccination coverage, which masks a real, negative density-case relationship that only becomes evident in the multivariate model. This finding argues against density as an independent predictor of measles transmission and reinforces vaccination rate as the primary structural factor.

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Spatial autocorrelation added a geographic dimension to this conclusion. Case counts were significantly clustered, but the specific pattern of that clustering is inconsistent with ordinary contagious spread across county lines. Berks County, despite direct adjacency to two outbreak counties, remains a statistical outlier in the cluster. This pattern is consistent with the possibility that the outbreak is influenced by the distribution of under-vaccinated communities in addition to geographic proximity. This is a pattern that geography alone cannot explain, but vaccination structure does.

The geographic analysis of Allegheny and Mifflin Counties refined this conclusion further, moving it away from a single structural signature of risk. Lancaster demonstrated that a concentrated, socially bounded under-vaccinated subgroup is sufficient to sustain a large outbreak once introduced. Allegheny demonstrated that this same structural feature is not significant on its own, an introduction event is a necessary complement, since a comparably large under-vaccinated pocket there has produced zero cases. This makes sense, as a measles infection cannot spontaneously manifest without some sort of transmission event. Mifflin demonstrated a second, distinct sufficient pathway entirely; a diffuse, county-wide vaccination gap with no internal subgroup contrast, still capable of sustaining a large outbreak once seeded. Together, these cases indicate that outbreak risk is not reducible to a single visible marker like a concentrated minority pocket but instead reflects a broader typology of susceptible-population structures that can each independently sustain transmission.

These conclusions carry an important caveat that should temper their applications. Every finding operates at the county or school level, describing population-level susceptibility structure rather than individual-level risk. The absence of sub-county case location data means that the link between school-level vaccination disparities and actual case origins remains a well-supported inference rather than a directly confirmed one. These are aggregate patterns, not claims about any specific child, family, or community.

From a surveillance standpoint, the practical implication is that visible structural outliers, like Lancaster's plain-community schools, should not be the sole focus of outbreak risk monitoring. Mifflin's diffuse, county-wide gap produced an outbreak just as readily as Lancaster's concentrated one, without any obvious structural signal to flag it in advance. Effective monitoring should therefore track both concentrated sub-county vaccination gaps and diffuse, county-wide under-vaccination as comparable risk profiles, rather than treating visible minority-community clustering as the primary or only warning indication.

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